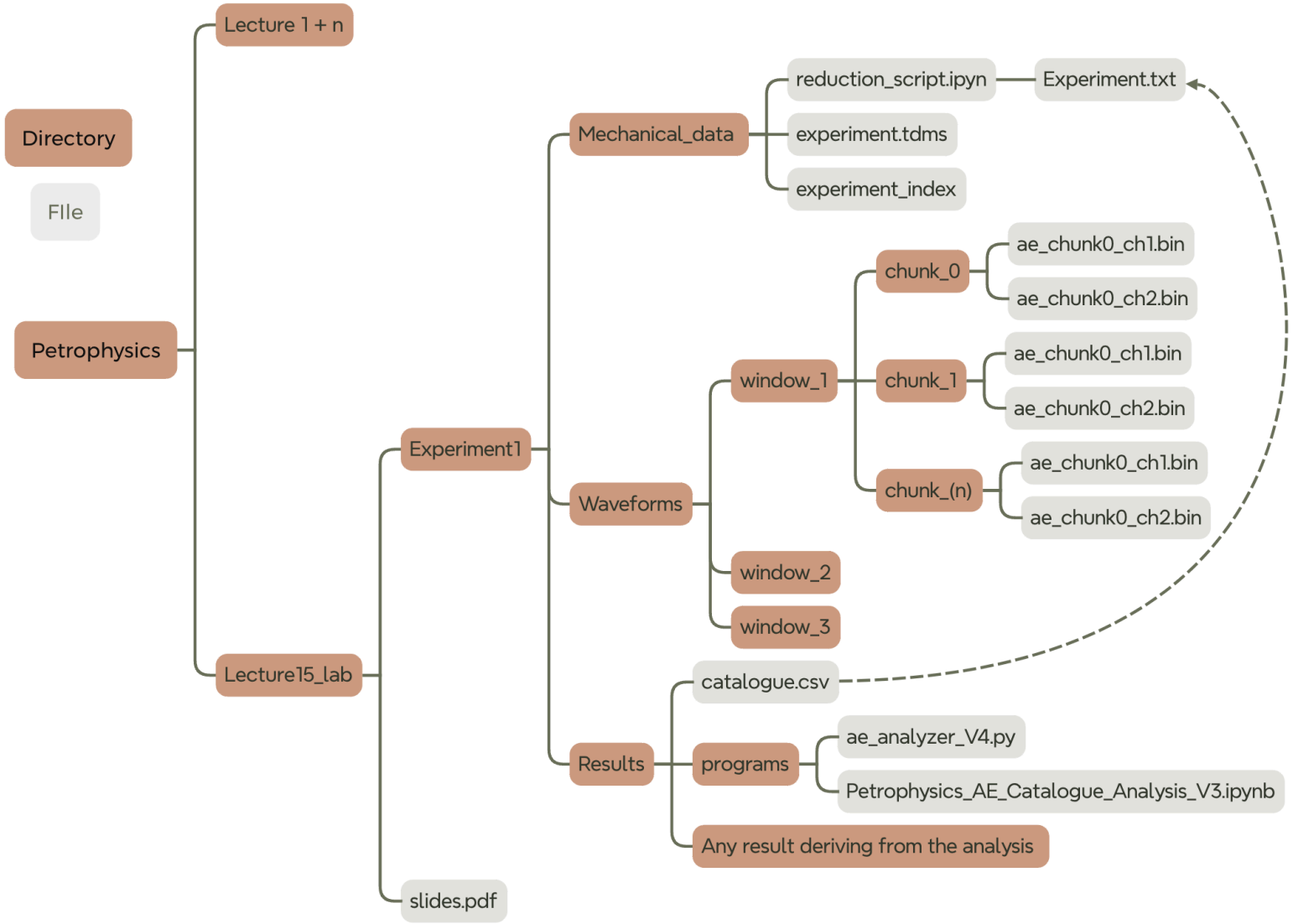
Data analysis workflow

Data structure



Main directories

Mechanical data

We store all the informations about the experiment and the files related to the mechanical data.

Waveforms directory

Here we put all the data related to the raw data waveforms to be analysed

Results directory (and subdirectories)

Here we store all the codes for the data analysis, the results from each analysis.

How to read the diagram

Brown boxes — directories (folders)

Grey boxes — files

Move into a folder

```
cd folder_name
```

Return to the parent folder

```
cd ..
```

Data analysis workflow

Two independent data streams meet through a shared experiment clock

MECHANICAL

Raw record

time, load, displacement

Reduced variables

stress, strain, slip

ACOUSTIC

Waveform files

window, chunk, channel

AE catalogue

onset, amplitude, quality

SYNCHRONIZE

Place every AE event on the mechanical history

Shared time coordinate
Recorded exposure retained
Windows remain distinct

OUTPUT

Mechanical curves + AE rate + amplitudes + rolling b-value proxy

The workflow preserves when data exist, where gaps occur, and which channel generated each detection.

Mechanical analysis

Question answered: what loading and deformation history did the sample experience?

1 Load

Read TXT, CSV, NPY or NPZ. Keep names and units.

2 Inspect

Check duration, sampling rate, columns and sign conventions.

3 Reduce

Apply calibration and compute stress, strain, displacement or slip.

4 Validate

Plot the full record and identify loading stages and failure.

Quality checks before interpretation

- **Time** monotonic and correctly sampled
- **Units** force, displacement and stress conversions
- **Offsets** sensor zero and discontinuities documented
- **Geometry** sample dimensions used consistently

MECHANICAL OUTPUT

One continuous experiment timeline

This timeline becomes the reference axis for every recording window and AE event.

Acoustic waveform analysis

Question answered: when did transient signals occur, and what properties did they have?

Data structure **window folder / ae_chunkN / ae_chunkN_chX.bin**

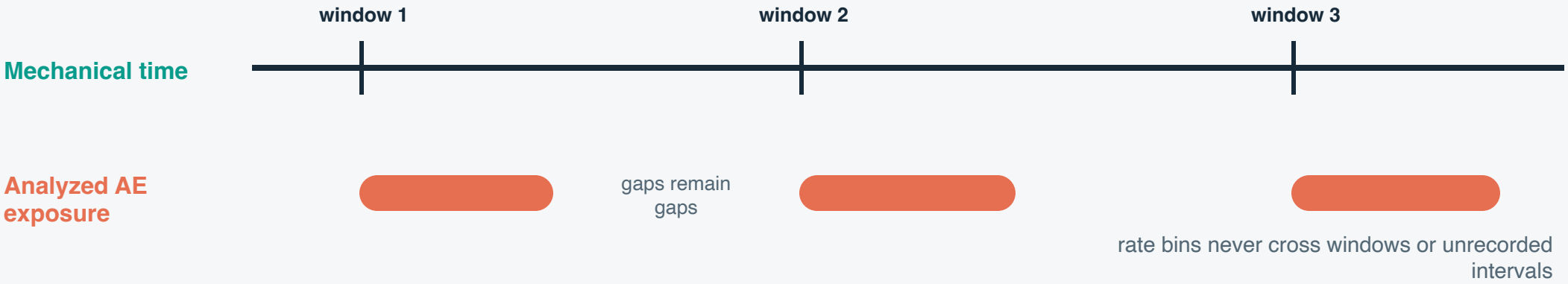
empty hardware channels are ignored

1. **Index files** — Associate each experiment window with its waveform directory.
2. **Read signal** — Detect dtype and sampling frequency, then load all valid samples.
3. **Prepare** — Optionally apply a band-pass filter before event detection.
4. **Detect** — Apply the STA/LTA picker and amplitude threshold consistently.
5. **Catalogue** — Store window, chunk, channel, onset time and peak-to-peak amplitude.

Output: one row per accepted AE detection

Mechanical and acoustic synchronization

Each recording window has its own anchor time, selected chunks and active channels



EVENT TIME

$$t_{event} = t_{window\ start} + chunk\ offset + onset\ in\ chunk$$

The catalogue records enough information to reconstruct the position of every event on the experiment timeline.

AE RATE

$$R = N_{accepted} / \Delta t_{exposure}$$

Calculate independently for every window and channel. Choose Δt to resolve the process without producing mostly empty bins.

Combined figure mechanical curve + AE rate + event amplitudes + recorded-window shading

Scientific interpretation

The combined dataset connects deformation, event occurrence and event-size evolution

Scientific question	Mechanical evidence	Acoustic evidence	Interpretation
When does damage begin?	Departure from linear behavior or strain change	First sustained increase in AE rate	Onset of distributed microcracking
Does failure approach gradually?	Stress evolution and proximity to peak or stress drop	Rate acceleration and shorter waiting times	Progressive damage or rapid instability
Does the event-size distribution change?	Loading stage and deformation state	Amplitude distribution and rolling b-value proxy	Relative shift toward larger detected events
Are observations robust?	Repeatable stages across the experiment	Agreement across windows and channels	Signal reflects sample behavior rather than coverage or one sensor

Interpret only after checking exposure, synchronization, picker consistency and channel coverage.